

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
ICSE CLASS 8 - PHYSICS

Chapter: Heat Transfer

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Heat naturally flows from:

- (i) A colder body to a hotter body
- (k) Both directions equally

- (j) A hotter body to a colder body
- (l) Only solids to liquids

Answer: _____

2. Evaporation mainly takes place:

- (i) Only at boiling point
- (k) Throughout the liquid only

- (j) At the surface of a liquid
- (l) Only below 0 degrees C

Answer: _____

3. Boiling differs from evaporation because boiling:

- (i) Occurs only at the surface
- (k) Can occur at any temperature

- (j) Occurs throughout the liquid at a fixed temperature
- (l) Needs no heat

Answer: _____

4. Heat is transferred mainly through a metal rod by:

- (i) Conduction
- (k) Radiation

- (j) Convection
- (l) Evaporation

Answer: _____

5. Heat transfer in liquids and gases mainly occurs by:

- (i) Conduction only
- (k) Reflection

- (j) Convection
- (l) Dispersion

Answer: _____

6. The Sun warms the Earth mainly by:

- (i) Conduction
- (k) Radiation

- (j) Convection
- (l) Evaporation

Answer: _____

7. A dull black surface is generally a:

- (i) Poor absorber and poor emitter
- (k) Good reflector only

- (j) Good absorber and good emitter
- (l) Perfect insulator

Answer: _____

8. Small gaps are left between railway tracks mainly to allow for:

- (i) Linear expansion
- (k) Cooling by convection

- (j) Superficial expansion only
- (l) Radiation loss

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Differentiate between evaporation and boiling using any two points. **[2]**

10. Explain how heat travels from the hot end to the cold end of a metal spoon. **[2]**

11. Explain briefly how convection currents are set up when water is heated from below. **[2]**

12. Explain the formation of a sea breeze during the daytime. **[2]**

13. What are linear, superficial and volume expansion? Give one example of any one of them. **[2]**

14. Why are cooking utensils made of metal but their handles are often made of wood or plastic? **[2]**
